MENUS

Full-Stack Web Based Food Delivery & Logistics Management Platform

Prepared by: AbdelRahman Khaled

TECHNICAL DOCUMENTATION & PORTFOLIO SPECIFICATION

1. EXECUTIVE SUMMARY

With the rapid expansion of digital commerce, modern online food delivery systems serve as crucial infrastructure connecting local restaurants, delivery personnel, and consumers. **MENUS** is a robust, full-stack web application designed to simplify online ordering while providing comprehensive administrative oversight. The system incorporates two-factor authentication (2FA) for administrative access, dynamic restaurant and menu navigation, order management, and operational logistics tracking.

2. KEY SYSTEM FEATURES

🔒 Authentication & Security Architecture

- **User Authentication:** Secure session handling and role-based login workflows for customers and administrators.
- **Two-Factor Authentication (2FA):** Enhanced security protocol requiring dynamic verification codes for administrative access.

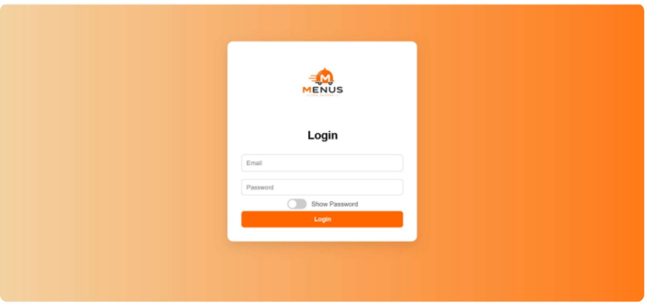

Figure 1.1: User & Admin Login Interface

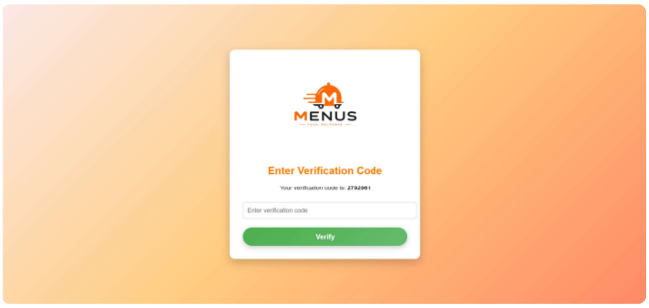

Figure 1.2: Two-Factor Authentication (2FA) Verification

🍔 Customer Ordering Portal

- **Multi-Restaurant Catalog:** Interactive browsing of local partner restaurants categorized by cuisine.
- **Dynamic Menu & Cart:** Live item selection, customized shopping cart management, and order submission with 30–40 minute delivery estimates.
- **Order History & Tracking:** Centralized dashboard for active order tracking and historical logs via `My Orders`.

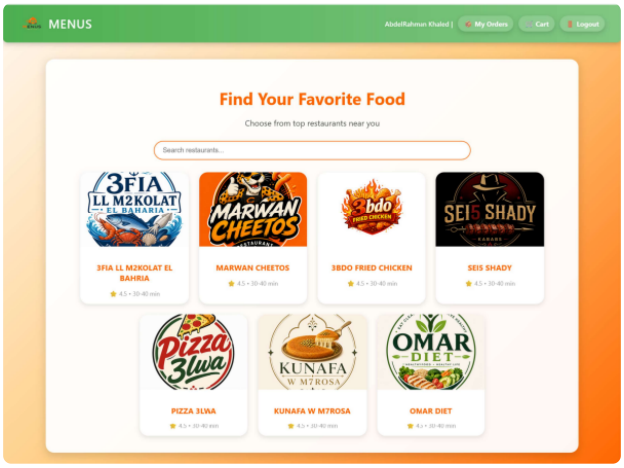

Figure 2.1: Restaurant Selection Portal & Catalog

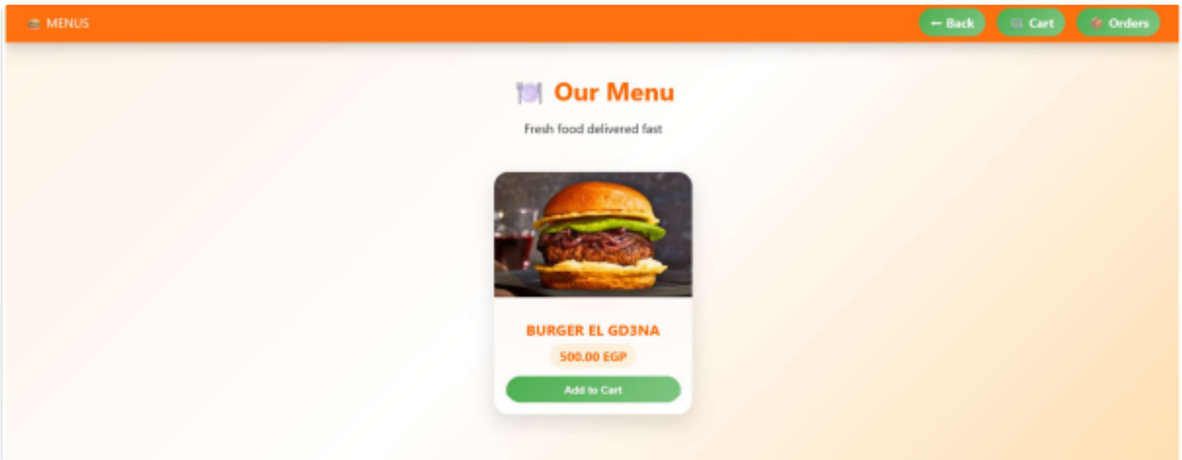

Figure 2.2: Dynamic Restaurant Menu & Item Ordering

🛠️ Administrative Management Dashboard

- **Restaurant & Menu Control:** Capabilities to create, update, or remove restaurant listings and itemized menus.
- **Logistics & Fleet Operations:** Driver dispatch management, task assignment, and real-time order status updates.

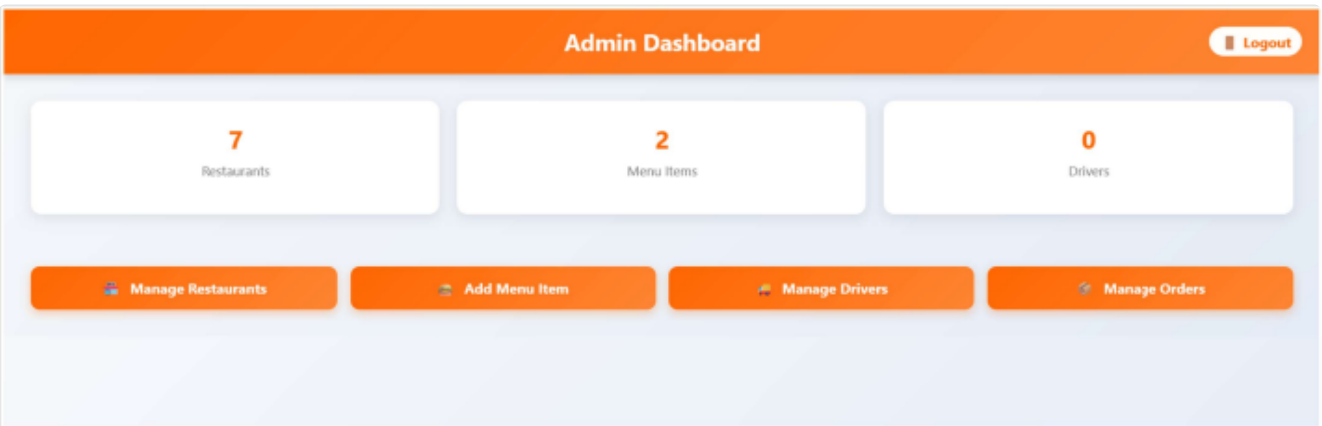

Figure 3.1: Administrative Metrics Overview & Action Center

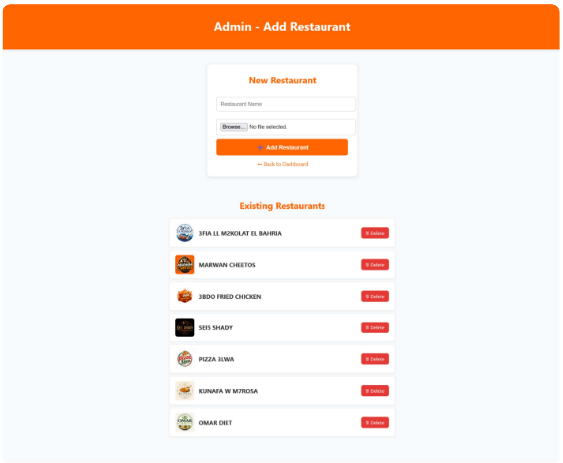

Figure 3.2: Restaurant Management & Catalog Maintenance

3. TECHNOLOGY STACK

FRONTEND	BACKEND	DATABASE	ENVIRONMENT
HTML5, CSS3	PHP	MySQL	XAMPP (Apache)

4. SYSTEM ARCHITECTURE & WORKFLOWS

Customer Workflow Flowchart

```
              +-----------------------+
              |   Index / Home Page   |
              +-----------+-----------+
                          |
       +------------------+------------------+
       |                  |                  |
+------v------+     +------v------+     +------v------+
| Guest User  |     | Register    |     | User Login  |
+------+------+     +------+------+     +------+------+
       |                  |                  |
+------v------+     +------v------+            |
| Guest Menu  |     | Account     |            |
| Preview     |     | Created     |            |
+-------------+     +-------------+            |
                                             |
       +-------------------------------------+
       |
+------v------+     +-------------+     +-------------+
| Restaurant  |---->| View Menu   |---->| Add to Cart |
| Catalog     |     +-------------+     +------+------+
+------+------+                              |
       |                                     |
       +-----------> My Orders <-------------+
                          |
                   +------v------+
                   | Checkout &  |
                   | Details     |
                   +-------------+
```

Admin Workflow Flowchart

```
+-------------------+
|    Admin Login    |
+---------+---------+
          |
+---------v---------+
|  2FA Verification |
+---------+---------+
          |
+---------v---------+
|  Admin Dashboard  |
+----+----+----+----+
     |    |    |    |
     |    |    |    +---------------------> Manage Orders
     |    |    |
     |    |    +--------------------------> Manage Drivers
     |    |
     |    +----> View / Add Restaurants
     |                 |
     |                 +-------------> Add Menu Items
     |
     +---------------------------------> System Analytics
```

5. DATABASE SCHEMA DESIGN

The platform relies on a normalized relational MySQL schema to ensure data integrity and query efficiency:

TABLE NAME	KEY ATTRIBUTES & DESCRIPTION
`customer`	Stores user profile metadata, encrypted login credentials, email, and primary shipping address.
`restaurant`	Contains establishment profiles, physical location data, and brand logo image references.
`menu`	Stores itemized menu offerings linked to target establishments via `restaurant_id` foreign key.
`drivers`	Maintains records of delivery personnel, contact telephone details, and assignment status.
`orders`	Tracks order headers, billing states, and customer links via `customer_id` foreign key.
`order_items`	Junction table managing order line items, quantity counts, and unit historical pricing.

6. FUTURE ENHANCEMENTS & ROADMAP

- → Integration of digital payment gateways (Stripe, PayPal, local card processing).
- → Real-time driver location tracking using interactive map APIs (Google Maps / Leaflet).
- → Customer rating and review engine for restaurants and individual food items.
- → Automated SMS and email notifications upon order state changes.